5.4.2 Important derivatives of cholesterol

Cholesterol is a precursor for other important steroid molecules such as the bile salts, steroid hormones, and vitamin D etc....

5.4.2.1 Bile salts

As polar derivatives of cholesterol, bile salts are highly effective detergents because they contain both polar and nonpolar regions. Bile salts are synthesized in the liver, stored and concentrated in the gall bladder, and then released into the small intestine. They help in the absorption of dietary forms of vitamin D in small intestine. Four bile acids are present in human bile, they are cholic acid, deoxycholic acid, chenodeoxycholic acid and lithocholic acid.

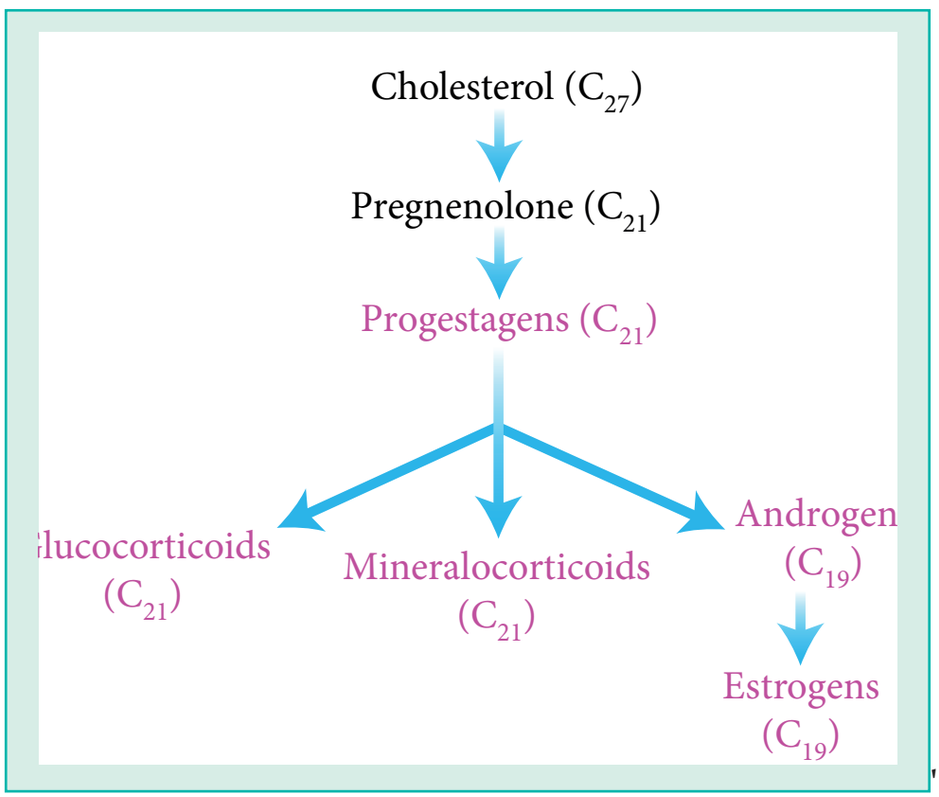
An efficient recycling system allows the bile salts to re-enter the bloodstream and return to the liver for reuse and only very small amount of bile salts escape this recycle system and this is the body's only route for cholesterol excretion.

5.4.2.2 Steroid hormones

Cholesterol is the precursor of the five major classes of steroid hormones: progestagen, glucocorticoids, mineralocorticoids, androgen, and estrogen.

5.4.2.3 Vitamin D

Cholesterol is also the precursor of vitamin D, which plays an essential role in the control of calcium and phosphorus metabolism. Vitamin D also serves as precursor for hormone synthesis.



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